FREESTANDING RESIDENTIAL DECK PLAN

Structural framing, footing layout, material schedule, cut list, and pricing snapshot

PROJECT DIMENSIONS

31'-0" x 21'-0"

LOCATION

Blue Springs, Missouri

FINISHED ELEVATION

Exactly 24" above finished grade

DESIGN BASIS

2024 IRC + local amendments

Key structural resolution. The fixed 24-inch deck height does not provide enough vertical room for conventional dropped triple-2x8 beams and the specified post hardware. This plan instead uses three flush triple-2x8 beams, 2x8 joists in hangers, 18 6x6 posts, and 22-inch pad footings.

Plan status. Construction-ready planning layout for municipal review; it is not an issued permit or an engineer's seal. Obtain the City's written permit determination and approval of the flush-beam, lateral, soil-bearing, and footing details before purchasing or building.

Code and pricing research checked July 15, 2026

1. Code, permit, and local requirements

Confirmed current Blue Springs requirements

- The City states that permit applications submitted on or after June 1, 2026 must comply with the **2024 International Residential Code (IRC)**. See the Blue Springs Codes Administration page.
- Ordinance 5480, adopted February 18, 2026, retains these local climatic values: **20 psf ground snow**, **115 mph wind**, Seismic Design Category A, severe weathering, **36-inch frost-line depth**, moderate-to-heavy termite hazard, and 6 degF winter design temperature. See Ordinance 5480, especially ordinance pages 16-17.
- The ordinance retains the permit exemption for uncovered platforms/decks/walks **not more than 30 inches above grade and not over a basement or story**. This 24-inch deck appears to satisfy that text. Ordinance page 22 and Chapter 500, §500.020 are the controlling municipal-code sources.
- The City's current Codes Administration webpage separately says construction on "a deck" requires a building permit. That conflicts with the ordinance's under-30-inch exemption and an older 2023 pamphlet. **Resolve the conflict in writing with Codes Administration, 816-228-0118, before work.** Do not rely only on the exemption.
- Zoning/setbacks still apply even if a building permit is waived. The City directs applicants to Planning at 816-228-0117 because setbacks vary by building type. An uncovered deck may encroach up to 30% into a required setback under the published pamphlet, but the parcel-specific setback and easements must be confirmed. See the Home Improvement Permit pamphlet.
- If a permit is required, the City accepts applications through its Permit & Application Center. The old deck pamphlet lists footing and final inspections; the current webpage says all permitted work receives applicable inspections and final approval. Ask the plan reviewer whether this flush-beam layout also needs framing inspection before decking. Inspection hotline: 816-228-0104.
- Inspection requests must be made by 3 p.m. the preceding day under the June 2026 update. Do not cover footings until the footing inspection is released.

Standard 2024 IRC criteria used

- Residential deck live load: **40 psf**; conventional wood-deck tables include **10 psf dead load** and wet-service factors.
- No guard is required where the walking surface is not more than 30 inches above grade measured within 36 inches horizontally of the edge. This plan has no guard because the stated grade is flat and the surface is 24 inches high. Recheck after final grading.
- 5/4 (actual approximately 1-inch) wood decking installed perpendicular to joists is permitted at up to 16 inches o.c.; nominal 2-inch wood decking is permitted at up to 24 inches o.c.
- Southern Pine No. 2 2x8 deck joists at 16 inches o.c. have an IRC table span of **11'-10"** at 40 psf live load. This plan's main clear span is only **9'-11/2"**.
- Beam and joist selections are also checked against the AWC DCA6 deck guide, which uses 40 psf live, 10 psf dead, No. 2 lumber, wet service, and L/360. DCA6 is supporting guidance, not a substitute for the adopted 2024 IRC.
- Beam-to-post gravity load must bear on the post through an approved post cap; a beam may not simply be bolted to the side of the post. See the American Wood Council explanation.

Conservative assumptions requiring City confirmation

- Soil bearing is assumed to be **1,500 psf** on undisturbed native soil. Confirm actual allowable bearing and the final 22-inch footing diameter with Blue Springs. Expansive fill, soft soil, groundwater, previous excavation, or uncontrolled fill invalidates the footing design.
- Footing bottoms are set at **38 inches below finished grade**, 2 inches below the confirmed 36-inch frost line. Confirm whether the inspector measures to bottom of pad and whether a belled pad/12-inch shaft detail is accepted.
- The freestanding lateral/uplift scheme-rated bases and caps, fully fastened joist hangers, corner knee braces, and underside diagonal straps-must be approved by the plan reviewer. The IRC does not supply one universal prescriptive lateral detail for every fully freestanding configuration.
- All structural lumber is No. 2 or better Southern Yellow Pine, pressure treated for exterior wet service. Use **UC4A/ground-contact treatment for posts and any member within 6 inches of soil**. Above-ground UC3B decking is acceptable where the product label permits deck use; ground-contact decking is also acceptable.

2. Final structural configuration

Item	Final design
Deck surface	31'-0" x 21'-0", exactly +24"
Decking direction	Parallel to 31-foot X axis
Joists	PT SYP No. 2, 2x8, actual 15-7/16" o.c. (never over 16")
Main joist spans	Two spans, each 9'-11/2" clear between beam faces
Edge outriggers	71/2" long, cut from joist offcuts; hanger at beam and 3 structural screws into rim
Joist cantilever	None; short outriggers are hanger-supported, not cantilevered
Beams	3 flush built-up beams, each (3) 2x8, 30'-101/2" long
Beam centerlines	Y = 1'-0", 10'-6", and 20'-0"
Posts	18 PT 6x6 posts; 6 per beam
Post X coordinates	1', 7', 13', 19', 25', 30'
Beam spans	6', 6', 6', 6', 5'; end overhangs 3" west and 111/4" east
Footings	18; 22" round x 8" pad at bottom, 12" shaft, bottom at -38", pier top +2"
Framing outside dimensions	30'-101/2" x 20'-101/2" before 3/4" fascia
Finished outside dimensions	31'-0" x 21'-0" at outside face of fascia/decking

Why 16-inch spacing is the best value

The 31-foot framing width is divided into 24 equal bays, so actual spacing is 370.5 ÷ 24 = **15.4375 inches**. A 12-inch layout needs 32 base joist lines plus extra framing at both end borders and at the breaker, roughly 36 lines rather than 28. It adds about 16 joist pieces, 48 hangers, fasteners, and labor. Because 2x8 joists span only 9'-11/2" (77% of their 11'-10" table limit), the chosen 16-inch layout already provides a modest stiffness upgrade. Twelve-inch spacing makes the surface slightly firmer but is poor value here.

Load and member checks

Design loads. Option A uses 40 psf live + 10 psf dead = **50 psf**. Option B is checked at 40 + 12 = **52 psf** to recognize the heavier 2x6 decking. The 20 psf local ground snow load is lower than the 40 psf live load and is not combined with live load.

Joist check, worst case (Option B). Tributary width at 16 inches = 1.333 ft. Uniform joist load = 52 x 1.333 = **69.3 plf**. At 9.125 ft clear span, simple-span moment = wL^2/8 = **721 lb-ft**. Actual span is 2'-81/2" shorter than the 11'-10" IRC table limit. Outriggers are only 71/2 inches and are supported by hangers, so no joist cantilever calculation is needed.

Interior beam. Its tributary width is 9.5 ft, so Option B line load = 52 x 9.5 = **494 plf**. For a 6-ft post spacing, M = wL^2/8 = **2,223 lb-ft** and end reaction = wL/2 = **1,482 lb**. A triple 2x8 has section modulus 3 x (1.5 x 7.25^2 ÷ 6) = **39.42 in^3**, requiring only 677 psi adjusted bending capacity. Shear demand is approximately 68 psi. Deflection using E = 1.26 million psi is approximately 0.08 inch, about L/900. These values are below normal wet-service No. 2 Southern Pine capacities; the City must nevertheless approve the built-up flush-beam calculation because the interior beam supports joists on both sides.

Outer beams. Tributary width = 1.0-ft perimeter strip + 4.75 ft = **5.75 ft**; Option B line load = 299 plf, substantially below the interior beam.

Posts and footings. Conservative interior-post tributary load = 494 plf x 6 ft = **2,964 lb**, plus a small beam/post self-weight allowance. A 22-inch round pad has area pi x 11^2 ÷ 144 = **2.64 ft^2** and nominal 1,500-psf bearing capacity = **3,960 lb**. The capacity-to-calculated-load ratio is about 1.30. Final diameter depends on actual soil and City approval.

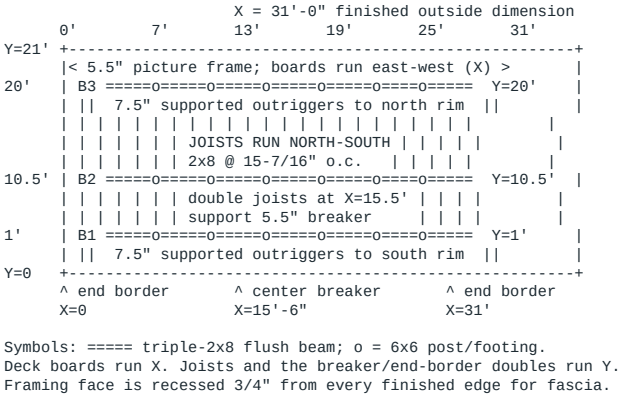
Dropped-beam clearance audit

Layer/elevation	5/4 decking	2x6 decking
Surface	24"	24"
Framing top	23"	221/2"
Bottom of 2x8 joist	153/4"	151/4"
Bottom of dropped 2x8 beam	81/2"	8"
Post bottom with pier +2" and 1" standoff	3"	3"
Remaining post height	**51/2"**	**5"**

The ABU66Z is approximately 6-1/16 inches tall and the BCS2-3/6Z post leg is approximately 2-15/16 inches. The requested dropped-beam assembly therefore cannot be installed with nonconflicting connector fasteners. In the flush design the post heights become 123/4 inches (Option A) and 121/4 inches (Option B), which fits.

3. Dimensioned framing and decking plan

Coordinate origin **(0,0)** is the southwest finished corner. X runs east along 31 feet; Y runs north along 21 feet. All footing coordinates are centerlines.



Symbols: ===== triple-2x8 flush beam; o = 6x6 post/footing.
Deck boards run X. Joists and the breaker/end-border doubles run Y.
Framing face is recessed 3/4" from every finished edge for fascia.

Major framing dimensions:

- Finished deck: 372" x 252"; diagonal for squaring = sqrt(372^2 + 252^2) = **449.33" = 37'-5 5/16"**. The other diagonal must match within 1/8 inch before concrete placement.
- Framing outside-to-outside before fascia: 370.5" x 250.5".
- Beam centers: Y = 12", 126", 240". Main clear joist spans: 123.75 - 14.25 = **109.5"** each.
- Joist centerlines: from X = 0.75" to 371.25" in 24 equal bays of 15.4375". Double the two end lines and the center-breaker line at X = 186".
- Outer rim inner face to outer-beam outer face: 9.75 - 2.25 = **7.5"** outrigger length.
- Add continuous solid blocking under the inner edge of each long picture-frame board, one midspan row in each main joist field, and bridge blocks between each doubled joist at not more than 24 inches o.c.

4. Footing coordinate schedule

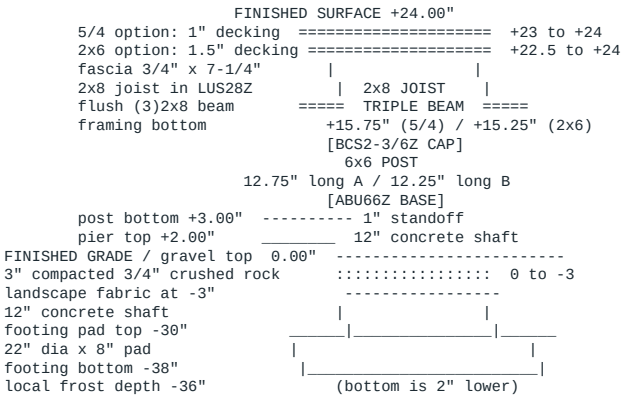
All footings are 22-inch round pads, 8 inches thick, with 12-inch shafts; bottom elevation -38 inches, top +2 inches. Adjacent X spacing is listed west-to-east.

ID	X	Y	Adjacent X spacing	Beam/post
F1	1'-0"	1'-0"	- / 6'	B1-P1
F2	7'-0"	1'-0"	6' / 6'	B1-P2
F3	13'-0"	1'-0"	6' / 6'	B1-P3
F4	19'-0"	1'-0"	6' / 6'	B1-P4
F5	25'-0"	1'-0"	6' / 5'	B1-P5
F6	30'-0"	1'-0"	5' / -	B1-P6
F7	1'-0"	10'-6"	- / 6'	B2-P1
F8	7'-0"	10'-6"	6' / 6'	B2-P2
F9	13'-0"	10'-6"	6' / 6'	B2-P3
F10	19'-0"	10'-6"	6' / 6'	B2-P4
F11	25'-0"	10'-6"	6' / 5'	B2-P5
F12	30'-0"	10'-6"	5' / -	B2-P6
F13	1'-0"	20'-0"	- / 6'	B3-P1
F14	7'-0"	20'-0"	6' / 6'	B3-P2

ID	X	Y	Adjacent X spacing	Beam/post
F15	13'-0"	20'-0"	6' / 6'	B3-P3
F16	19'-0"	20'-0"	6' / 6'	B3-P4
F17	25'-0"	20'-0"	6' / 5'	B3-P5
F18	30'-0"	20'-0"	5' / -	B3-P6

Y spacing between beam/footing rows is 9'-6" and 9'-6". Pad edges remain at least 1 inch inside the finished footprint at the closest edges.

5. Side section and exact elevations



Excavate approximately 3 inches beneath the deck and return with compacted gravel so the final gravel top equals the original finished-grade datum. Do not bury the post base or concrete top.

6. Decking layouts - both options

Both options use actual 51/2-inch board width. Verify every bundle before framing; if the measured average differs, redo the course calculation.

Common exact layout

- End picture-frame boards: one 51/2-inch course at X = 0-5.5" and one at X = 366.5-372".
- Center breaker: one 51/2-inch board centered at X = 186", occupying X = 183.25-188.75".
- Provide four 1/8-inch longitudinal separation gaps: end border/field and field/breaker on each side.
- Each field-board cut length = (372 - 3x5.5 - 4x0.125) ÷ 2 = **177.5" = 14'-91/2"**. Every field piece comes from 16-foot stock; no random field butt joints.
- Across the 21-foot width use **43 full field courses** between two 51/2-inch long picture-frame borders. Board width total = 45 x 5.5 = 247.5 inches. Remaining gap total = 252 - 247.5 = 4.5 inches over 44 spaces, average **0.1023 inch**.
- Practical balanced story pole: use 32 gaps at 3/32 inch and 12 gaps at 1/8 inch, distributed symmetrically (for example, insert a 1/8 gap about every fourth space and mirror the sequence from center). Total = 32x3/32 + 12x1/8 = **4.5 inches exactly**. Measure actual boards first; wet boards may need smaller installation gaps to reach the same final total.
- Set the outer fascia/decking control lines first. Install field courses from both long sides toward the center and use a story pole; do not rip a narrow final board. The calculation yields 43 full-width courses.
- Long picture-frame sides fit between the two full-width end borders: net 361 inches. Use two equal 180-7/16-inch pieces with a 1/8-inch center joint on solid blocking.
- Each 21-foot end border: 12-foot piece + 107-7/8-inch piece + 1/8-inch joint = 252 inches.
- Center breaker between long borders: 2403/4 inches; use 12-foot piece + 96-5/8-inch piece + 1/8-inch joint.

Option A - 5/4x6 decking

- Install crown/bark side consistently; two #9 x 21/2-inch ACQ-rated exterior deck screws at every supporting joist/block, approximately 3/4 inch from each board edge.

- Predrill all fasteners within 1 inch of an end, every screw in picture-frame/breaker pieces, and any hole near a knot. Use a bit approximately 1/64 inch smaller than the screw root.
- Minimum screws calculated approximately 2,260; buy **2,500** (10%+ allowance).
- Best value: lighter, easier to handle, permitted at the selected spacing, and currently the lowest verified decking cost.

Option B - 2x6 decking

- Same board count and layout; use #9 or #10 x **3-inch** ACQ-rated deck screws, two at every support.
- Predrilling requirements are the same; predrill every board end because the thicker material is more likely to split near butt joints.
- Buy **2,500** screws.
- Structural change: post cuts are 121/4 inches instead of 123/4 inches. Footings, beams, joists, blocking, posts, and connectors remain unchanged because they were checked at 52 psf.

For both options use 305 stainless screws where the deck will receive deicing salt or frequent chemical exposure; otherwise use a manufacturer-labeled polymer/triple-coated or hot-dip-galvanized fastener approved for the exact treatment chemical. Never mix stainless fasteners with incompatible plated connectors in direct wet contact.

7. Fascia and cut-end treatment

- Fascia: 3/4-inch actual x 71/4-inch PT 1x8, covering all four rim faces and the exposed ends of the three beams. Use eight 16-foot boards including overage.
- Keep fascia top flush with framing top and decking/fascia outside faces on the 31x21 control lines. Use 45-degree outside corners or square-butt corners with the meeting piece shortened by twice the 3/4-inch thickness; do not add dimensions twice.
- Leave 1/8 inch at intermediate fascia butt joints and approximately 1/16 inch at dry-weather miter joints. Provide solid blocking behind every joint.
- Fasten with two vertical rows of #9 x 21/2-inch ACQ-rated fascia/trim screws at 24 inches o.c.; 120 screws including allowance. Predrill fascia ends.
- Brush two flood coats of **copper naphthenate end-cut preservative** meeting AWPA M4 on every field cut, notch, and drilled end grain before assembly. One gallon is ample. Do not use untreated ripped stock as exterior blocking.

8. Ground preparation

- Exact covered area: 31 x 21 = **651 ft^2**.
- Fabric purchase: 651 x 1.10 = **716.1 ft^2**; buy at least 800 ft^2 to permit 12-inch overlaps and pier cuts.
- Compacted gravel at 3 inches: 651 x 0.25 ÷ 27 = **6.03 yd^3 compacted**.
- Add 10% waste plus about 10% loose-to-compacted allowance: 6.03 x 1.20 = **7.24 yd^3**; order **7.4 yd^3**, approximately **10.4 tons** at 1.4 tons/yd^3. Confirm the quarry's actual density.
- Use clean, angular **3/4-inch crushed limestone or crushed rock with limited fines**, not pea gravel. It locks together and drains while discouraging vegetation.

Sequence: remove sod/organic material; excavate approximately 3 inches while maintaining positive drainage; pour/inspect footings; fine-grade soil to drain at least 1/8 inch per foot toward a safe outlet and never toward the house; compact subgrade; install nonwoven landscape fabric with 12-inch overlaps and seal/cut neatly around piers; place gravel in two lifts; compact each lift; restore the top to the original finished-grade datum. Keep gravel below the concrete tops and out of post bases. Low-profile metal/plastic edging at the perimeter is optional where gravel migration is likely; do not create a dam that traps water.

9. Hardware and fastener schedule

All connectors and fasteners contacting treated lumber must be ZMAX/G185 hot-dip galvanized, mechanically galvanized as specifically approved, or Type 304/305/316 stainless. Use one compatible metal system.

Item	Required quantity	Example / installation
6x6 adjustable standoff bases	18	Simpson ABU66Z, ZMAX; 1" standoff
5/8" cast-in anchors	20 (18 + 2 spare)	HDG 5/8" x 10" J/anchor bolts, centered; template-set
Triple-2x beam/6x6 caps	18	Simpson BCS2-3/6Z, ZMAX

Item	Required quantity	Example / installation
2x8 joist hangers	170 (168 + 2 spare)	Simpson LUS28Z; 112 main-joist ends + 56 outer outriggers
BCS connector screws	200	Simpson SD10112, installed only in the manufacturer schedule
ABU nails	250	0.148" x 11/2" HDG connector nails; use exact ABU schedule
ABU through-bolt sets	40	1/2" HDG bolts, nuts, washers; two per base if required by selected table
Hanger header nails	500	0.148" x 3" HDG connector nails
Hanger joist nails	750	0.148" x 11/2" HDG connector nails
Built-up beam screws	225	Simpson SDW22300/approved 3" structural screws; two staggered rows at 24" o.c. on each outer face
Blocking/rim screws	800	#10 x 3" ACQ-rated structural exterior screws; 3 per block end, 3 through rim into each outrigger
Knee-brace bolts	20 sets	1/2" x 8" HDG through-bolts, washers/nuts; one per brace end (two per brace), plus spares, as detailed/approved
Underside diagonal strap	1 x 100-ft roll	Simpson CS16-R or approved; two diagonal runs; fasten per manufacturer
Deck screws, Option A	2,500	#9 x 21/2" ACQ-rated coated or 305 SS
Deck screws, Option B	2,500	#9/#10 x 3" ACQ-rated coated or 305 SS
Fascia screws	120	#9 x 21/2" ACQ-rated trim/fascia screws
End-cut treatment	1 gal	Copper naphthenate meeting AWPA M4
Optional top tape	1 lot	2" joist tape and 6" beam tape; useful but not structurally required

Do not substitute ordinary deck screws for connector nails, structural screws, bolts, or hanger fasteners. Follow the current connector load table; the fastener counts above are purchasing quantities, not permission to alter a listed fastening schedule.

10. Material schedules

Common structural/ground package - both options

Quantities already include approximately 10% overage where practical.

Material	Purchase quantity	Calculation/use
PT SYP No. 2 2x8x10	70	56 main joists + 13 blocking boards + 1 spare
PT SYP No. 2 2x8x8	10	9 beam segments + 1 spare
PT SYP No. 2 2x8x12	17	15 beam segments + 2 spare
PT SYP No. 2 2x8x16	8	4 perimeter-rim halves + 3 long beam segments + 1 spare
PT 6x6x8 UC4A	4	3 minimum boards yield 18 short posts; one overage board
PT 2x6x8	3	Eight 24" corner braces; one spare board
PT 1x8x16 fascia	8	104 perimeter ft plus joints/overage
80-lb 4,000-psi concrete	128 bags	18 x 3.85 ft^3 ÷ 0.60 = 116; rounded with 10% allowance
12" x 48" fiber forms	18	One 32" shaft form per footing
Landscape fabric	800 ft^2	651 ft^2 + overlaps/10%
3/4" crushed rock	7.4 yd^3 / ~10.4 tons	3" compacted + waste/compaction
Hardware/fasteners	Per §9	No duplication with decking schedules

Required footing excavation is larger than the concrete pad so inspectors can see undisturbed bearing soil and pad dimensions. Backfill only after release.

Option A - 5/4x6 decking package

Piece	Purchase quantity	Calculation
5/4x6x16 PT SYP	100	86 field pieces + 4 long-border pieces = 90; x1.10 = 99
5/4x6x12 PT SYP	4	3 exact for two end borders + breaker; one overage
5/4x6x10 PT SYP	4	3 exact short border/breaker tails; one overage
#9 x 21/2" deck screws	2,500	Calculated ~2,260 + allowance

Option B - 2x6 decking package

Piece	Purchase quantity	Calculation
2x6x16 PT SYP	100	Same 90-piece exact layout x1.10
2x6x12 PT SYP	4	End borders/breaker + overage
2x6x10 PT SYP	4	End borders/breaker + overage
#9/#10 x 3" deck screws	2,500	Calculated ~2,260 + allowance

Optional upgrades, excluded from required totals: stainless deck screws; joist/beam top tape; #1/premium appearance decking; rebar if required by the City/soil condition; low-profile gravel edging.

11. Shop-ready cut lists

Common framing cut list

Material / stock	Finished cuts	Required	Offcut and reuse
2x8x10	9'-11/2" main joist	56	101/2" offcut; cut one 71/2" outer outrigger from each, leaving ~3"
2x8x10	13-15/16" bay block	96	Eight per board; about 81/2" waste per stock board
2x8x10	4" bridge block between doubles	30	One board yields 30 before kerf; use a second/spare if kerf accumulates
2x8x8	6'-3" beam segment	6	1'-9" offcut; usable for approved blocking only
2x8x8	5'-111/4" beam segment	3	2'-03/4" offcut; usable blocking
2x8x12	12'-0" beam segment	9	No offcut
2x8x12	11'-111/4" beam segment	6	3/4" trim
2x8x16	12'-3" beam segment	3	3'-9" offcut; blocking/bracing stock
2x8x16	15'-5 3/16" rim half	4	~6-13/16" offcut; two halves + 1/8" joint = 30'-101/2"
6x6x8	123/4" post (A) / 121/4" post (B)	18	Three stock boards suffice; fourth is defect/overage stock
2x6x8	24" knee brace, ends field-fit	8	Two stock boards exact before diagonal trims; third is spare
1x8x16 fascia	field-cut miter/butt segments	8 boards	Keep outside-to-outside fascia exactly 31' x 21'; back every joint

Beam assembly per beam: outer plies use 6'-3" + 12'-0" + 11'-111/4" with splices centered over posts at X=7' and 19'. Center ply uses 12'-3" + 12'-0" + 5'-111/4" with splices at X=13' and 25'. Every splice is directly over a post with full bearing; never splice between posts.

Option A decking cut list

Stock	Finished cut	Required	Offcut/reuse
5/4x6x16	14'-91/2" field piece	86	141/2" each; nonstructural scrap or small trim
5/4x6x16	15'-0 7/16" long-border half	4	11-9/16" each
5/4x6x12	12'-0"	3	None
5/4x6x10	8'-117/8" end-border tail	2	1'-01/8" each
5/4x6x10	8'-05/8" breaker tail	1	1'-113/8"
Overages	Keep uncut until final board-width verification	10x16', 1x12', 1x10' nominal allowance	Replace defects and measurement variation

Option B decking cut list

Use the identical lengths and counts above in 2x6 stock. The only cut-list changes are the board thickness, 3-inch deck screws, and 121/4-inch post cuts.

12. Construction sequence

1. **Verify authority.** Obtain written permit determination, parcel setbacks/easements, HOA approval, soil/footing acceptance, and lateral-detail acceptance. Contact Missouri 811 at least three working days before excavation; private lines may require a separate locator. Never excavate until every utility has responded.
2. **Establish footprint.** Set offset batter boards beyond the 31x21 rectangle. Protect a benchmark representing finished grade 0.00 and a laser benchmark at deck surface +24.00.
3. **Square it.** Set X strings 31'-0" and Y strings 21'-0". Adjust until both diagonals are 37'-5 5/16" and equal within 1/8". Recheck after every string movement.
4. **Mark footings.** Plumb down all 18 coordinates from crossing strings. Mark 22-inch pads and larger working excavations. A common error is measuring from framing rather than the finished (0,0) corner.
5. **Excavate.** Dig to undisturbed bearing at least 38 inches below finished grade. Hand-dig in utility tolerance zones. Protect open holes from falls, rain, and collapse.
6. **Form/pour.** Form an 8-inch-thick, 22-inch round pad from -38 to -30 and a centered 12-inch shaft from -30 to +2. Use 4,000-psi bag mix at the labeled water ratio; do not add water for workability. Place monolithically where practical, consolidate, and keep soil out. **Inspection:** footing excavation/reinforcement (if required) before concrete.
7. **Set anchors/bases.** Template-set centered 5/8-inch HDG anchors while concrete is plastic, or use an approved post-installed system only after cure. After specified cure, install ABU66Z bases level and aligned; do not use bases to compensate for badly misplaced piers.
8. **Posts.** Laser the three beam-bottom elevations. Cut/treat posts to 123/4 inches (A) or 121/4 inches (B), but field-measure every pier and trim individually so beam bottoms land exactly at +153/4/+151/4. Do not assume all pier tops are identical.
9. **Build beams.** Crown plies upward, stagger splices over specified posts, clamp straight, and install two staggered rows of SDW screws at 24 inches o.c. from each outer face. Keep ends at X=.75" and 371.25". Set in BCS caps and install only approved connector fasteners.
10. **Joists/rims/blocking/bracing.** Lay 24 equal 15-7/16-inch bays. Hang 56 main joists; install 56 offcut outriggers at outside faces of B1/B3; install outer rims; double end-border and center-breaker lines; install 96 bay blocks and bridge blocks. Treat cuts before fastening.
11. **Check geometry/elevation.** Verify 370.5x250.5 framing, square diagonals, beam/post plumb, and framing-top elevations +23 (A) or +221/2 (B). Plane only minor high crowns; replace badly crowned/twisted pieces.
12. **Ground work.** After footing release and before decking blocks access, excavate/grade the underdeck area. Keep soil away from bases.
13. **Fabric/gravel.** Install 800 ft^2 fabric with 12-inch laps, then compact 7.4 yd^3 loose rock in two lifts to final grade 0.00. Maintain drainage.
14. **Picture-frame/breaker support.** Verify solid support beneath every board edge, end, butt joint, and the four 1/8-inch longitudinal gaps. The center and end strips require doubled framing; long borders require a continuous inner blocking row.
15. **Dry layout.** Measure 10-20 boards face-to-face to determine average actual width. Mark the 43-course story pole and four longitudinal field limits. Recalculate if average width is not 51/2 inches.
16. **Field decking.** Install 86 pieces at 14'-91/2", working symmetrically. Use two screws at each support. Snap lines so screw rows remain straight; do not force bowed boards by widening the deck.
17. **Breaker.** Install the 12-foot and 8'-05/8" pieces with a 1/8-inch supported joint, centered at X=15'-6". Predrill.
18. **Picture frame.** Install end borders and long borders to the fixed 31x21 outside control lines. Measure outside-to-outside before final screws. A common error is adding fascia thickness after already setting a 31x21 frame.
19. **Fascia.** Install 3/4x71/4 fascia flush with framing top; outside faces define the exact finished rectangle. Two rows of fascia screws at 24 inches o.c.; back all joints.
20. **Final checks.** Verify every connector hole required by its schedule is filled, anchors/bolts are tight, no untreated cut is exposed, drainage is open, surface is +24, dimensions are 31x21, and no edge exceeds the guard threshold after grading. Prepare footing photos, product labels, connector schedules, and this coordinate plan for final inspection.

Safety: use trench/open-hole barriers, eye/hearing/respiratory protection, gloves for treated lumber and concrete, GFCI-protected tools, proper lifting for wet lumber, and temporary bracing until the deck is self-supporting. Never work under an unsupported beam.

13. Materials-only pricing snapshot

Pricing limitations and availability findings

Prices were checked July 15, 2026 and are **pre-tax planning figures**, excluding permit, labor, delivery, tools, and rentals. Retail sites change prices by store/session.

- **Home Depot:** public pages showed current product and quantity-break prices for qualifying #2 Prime ground-contact SYP 2x8 and 5/4 decking. For example, 2x8 SYP product family and 5/4x6 decking. Exact Blue Springs pickup inventory was not exposed reliably; verify carts before ordering.
- **Lowe's:** the closest localized store surfaced as Independence (64055). The exact qualifying ground-contact 5/4x6x16 and 2x6x16 SYP listings both displayed **OUT OF STOCK** and no sale price. Lowe's therefore cannot supply a complete qualifying basket today; inventing a total or substituting another retailer's price would violate the brief. Product links: 5/4 listing, 2x6 listing, and ABU66Z. A complete Lowe's estimate is **not currently bid-able**.
- **Menards:** public pages showed everyday prices and in-store/special-order status but required a store check for actual pickup quantities. No expired 11% rebate is deducted. Qualifying references include AC2 ground-contact SYP, 2x8x14 AC2, and 5/4 premium decking.
- Gravel is priced as 400 half-cubic-foot retail bags so each store basket is comparable. A local quarry's 10.4-ton bulk quote will usually be far cheaper even after delivery, but quarry pricing is outside the three required retailer baskets.

Retailer estimate summary

These totals use the full overage quantities in §§9-10. "Allowance" denotes a same-retailer product group whose exact local SKU price was not exposed; verify before purchase.

Category	Home Depot A 5/4	Home Depot B 2x6	Menards A 5/4 premium	Menards B 2x6	Lowe's A/B
Structural framing	$1,845	$1,845	$2,480	$2,480	N/A-decking/framing stock incomplete
Decking	$1,363	$1,816	$3,420	$2,525	N/A-qualifying 16' products out of stock
Concrete/forms/anchors	$1,203	$1,203	$1,033	$1,033	Incomplete
Hardware/fasteners	$2,791	$2,821	$2,556	$2,586	Incomplete
Fascia	$144	$144	$233	$233	Incomplete
Fabric/gravel	$2,352	$2,352	$1,696	$1,696	Incomplete
Complete pre-tax total	**$9,698**	**$10,181**	**$11,418**	**$10,553**	**Not currently purchasable as specified**

Representative priced lines behind the estimate:

Retailer / product	Qty	Unit used	Extended	Status/link
Home Depot 2x8x10 #2 Prime GC SYP	70	$15.55 quantity price	$1,088.50	Product family above; verify local
Home Depot 2x8x8 #2 Prime GC SYP	10	$11.59 quantity price	$115.90	2x8 family
Home Depot 2x8x12 #2 Prime GC SYP	17	$17.71 quantity price	$301.07	Same family
Home Depot 2x8x16 #2 Prime GC SYP	8	$24.01 quantity price	$192.08	Same family
Home Depot 5/4x6x16 standard GC SYP	100	$12.76 quantity price	$1,276.00	Decking family
Home Depot ABU66Z	18	$51.34 quantity price	$924.12	ABU66Z
Menards 5/4x6x16 #1 premium GC SYP	100	$32.49 everyday	$3,249.00	AC2 premium decking
Menards 1x8x12 premium GC SYP fascia equivalent	12-equivalent allowance	$19.39	$232.68	1x8 product
Lowe's 5/4x6x16 standard GC SYP	100 required	No price	N/A	Out of stock at localized store
Lowe's 2x6x16 #2 Prime GC SYP	100 required	No price	N/A	Out of stock at localized store

Lowest-cost combined basket: Home Depot framing, fascia, and decking; Menards concrete/forms, hardware/fastener allowance, fabric, and gravel. Estimated total **$8,637 Option A** or **$9,120 Option B**. Every combined-basket category and supplier is identified in the preceding sentence; validate product grade, treatment, and checkout quantity before buying. A one-retailer Home Depot bundle is

only about $1,061 higher for Option A and may become cheaper after an unlisted job-lot discount or avoided split logistics, even though delivery is excluded here.

The totals are deliberately not presented as quotes. Local inventory for 100 straight 16-foot boards is the largest risk; arrange a contractor desk takeoff, inspect lumber, and get a written job-lot quote from each retailer.

14. Final comparison and recommendation

Choice	Cost/weight	Stiffness/installation	Recommendation
5/4x6	Lowest verified cost; lighter	Code-permitted at 16"; easy handling; 21/2" screws	**Best value**
2x6	About $480 more in the HD basket; roughly 50% thicker/heavier	Firmer board feel, 3" screws, harder to straighten; framing unchanged	Choose only if thicker appearance/impact resistance is worth cost
16" o.c.	28 framed lines	Main joists at only 77% of table span; good stiffness	**Best value**
12" o.c.	Roughly 36 lines; more lumber and ~48 additional hangers	Firmer surface, especially as decking weathers	Optional upgrade, not economical here

Recommended final configuration: 5/4x6 PT SYP decking; 2x8 PT SYP No. 2 joists at actual 15-7/16 inches o.c.; three flush triple-2x8 beams; 18 short 6x6 UC4A posts; 22-inch pads with 12-inch shafts to 38 inches below grade; full picture frame; one center breaker; face-driven ACQ-rated screws. This balances common stock, modest stiffness reserve, uncomplicated geometry, cost, and long-term serviceability.

15. Consistency audit and stop-work conditions

- Geometry: 31x21 finished dimensions, 30'-101/2" framing, 3/4" fascia, 43 field courses, 86 field pieces, 18 footings, 3 beams, 6 posts/beam, and all coordinates agree.
- Elevation: +24 surface; +23/+221/2 framing top; +153/4/+151/4 framing bottom; +3 post bottom; 123/4/121/4 posts. Flush beams fit; dropped beams do not.
- Loads: Option B's 52 psf governs joists, beams, posts, and footings. The 20 psf local snow load is lower than live load.
- Concrete: 22" x 8" pad plus 12" x 32" shaft is about 3.85 ft^3 per footing; 18 footings need about 116 80-lb bags, and 128 includes about 10%.
- Gravel: 651 ft^2 x 3 inches = 6.03 compacted yd^3; 7.4 yd^3 purchase includes waste/compaction.
- The largest material-price uncertainty is retailer-local stock and pricing; Lowe's was confirmed incomplete, and Home Depot/Menards quantities require store verification.

Mandatory cautions: underground utilities must be located before excavation. Final footing diameter/depth depends on verified soil and Blue Springs approval. The Blue Springs building department must approve the final design or confirm exemption in writing. A Missouri-licensed structural engineer must review any condition outside prescriptive deck provisions, including poor soil, unusual wind exposure, altered elevations, roofs, hot tubs, heavy equipment, or a rejected lateral detail. Verify lumber species, grade stamp, treatment category, retention, straightness, moisture, and actual dimensions before purchase. Retail prices and inventory change. Do not use this plan if field conditions differ materially from the stated flat, unobstructed, normal-residential assumptions.